Computer Science - Undergraduate Level Assessment 9 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. In Python 3, which of the following function removes all leading and trailing whitespace in string?
   (a) replace(old, new [, max])
   (b) strip([chars])
   (c) swapcase()
   (d) title()

2. Which of the following regular expressions is equivalent to (describes the same set of strings as) (a* + b)*(c + d)?
   (a) a*(c + d)+ b(c + d)
   (b) a*(c + d)* + b(c + d)*
   (c) a*(c + d)+ b*(c + d)
   (d) (a + b)*c +(a + b)*d

3. In Python 3, what is the output of print list[1:3] if list = ['abcd', 786 , 2.23, 'john', 70.2]?

786 , 2.23, 'john', 70.2

   (a) abcd

786, 2.23

   (b) None of the above.

4. What is the output of the statement "a" + "ab" in Python 3?
   (a) Error
   (b) aab
   (c) ab
   (d) a ab

5. Which of the following best explains what happens when a new device is connected to the Internet?
   (a) A device driver is assigned to the device.
   (b) An Internet Protocol (IP) address is assigned to the device.
   (c) A packet number is assigned to the device.
   (d) A Web site is assigned to the device.

True / False (10 marks)

Answer True or False

6. True or False: The set of Boolean operators {AND, OR} is not complete for representing all Boolean expressions.
7. True or False: The access matrix approach can express complex protection requirements without limitations.
8. True or False: A distributed denial-of-service (DDoS) attack involves multiple computers launching the attack simultaneously.
9. True or False: The maximum possible degree of the minimal-degree interpolating polynomial p(x) for n + 1 distinct points is n + 1.
10. True or False: A function that sorts values in a column or row is useful for identifying outliers in a dataset caused by data entry errors.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to extract the maximum numeric value from a string by using regex.
12. Write a function to find the minimum total path sum in the given triangle.

End of Paper